CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

Salesforce Implementation Project Documentation

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1. Project Overview

Customer support teams receive many tickets every day, and not every issue needs the same level of attention. When tickets are checked and assigned manually, urgent issues can take longer to reach the right person. This project introduces a Salesforce-based solution that uses Agentforce and Flow to analyze support ticket descriptions, identify their priority, and support automated assignment.

The system classifies tickets into High, Medium, or Low priority. For high-priority cases, it can create an urgent handling task and assign the ticket to a senior support agent. This helps reduce manual work and gives the support team a clearer way to handle urgent cases.

2. Problem Statement

Customer support teams may receive a large number of tickets each day. If the tickets are prioritized and assigned manually, important issues may not be handled quickly. The project therefore focuses on creating a simple intelligent workflow that can understand the ticket description, identify the urgency, and start the appropriate automation.

- Automatically classify ticket priority.
- Assign tickets to the appropriate support level.
- Reduce manual work for support teams.
- Help urgent issues receive faster attention.
- Use Agentforce to analyze ticket information and trigger backend automation.

3. Objectives

- To create a custom Salesforce object for storing support ticket information.
- To capture important details such as customer, contact, issue type, description, priority, status, and assigned agent.
- To analyze ticket descriptions using predefined urgency-related keywords.
- To automatically classify tickets as High, Medium, or Low priority.
- To create an urgent handling task for High-priority tickets.
- To connect Agentforce with Flow so that the process requires less manual intervention.

4. User Stories

User	Requirement
Support Agent	Wants tickets to be prioritized automatically so critical issues can be handled first.
Customer	Wants urgent issues to be resolved quickly for a better support experience.

Manager	Wants tickets to be assigned to the right team so workload can be managed.
AI Agent (Agentforce)	Analyzes ticket details and triggers automation to reduce manual work.

5. Expected Outcome

- Faster handling of urgent support tickets.
- Improved customer support experience.
- Reduced manual effort.
- More organized ticket handling.
- Better productivity for the support team.

6. Requirement Analysis

6.1 Functional Requirements

- The system should store customer support ticket information.
- The system should retrieve the latest ticket associated with an account.
- The system should analyze the ticket description.
- The system should assign High, Medium, or Low priority.
- The system should create a Task for High-priority tickets.
- The system should provide the priority and action information back to Agentforce.

6.2 Non-Functional Requirements

- The process should be simple for support users.
- The automation should provide consistent results for the configured keyword rules.
- The system should reduce unnecessary manual steps.
- The configuration should be easy to maintain and extend.

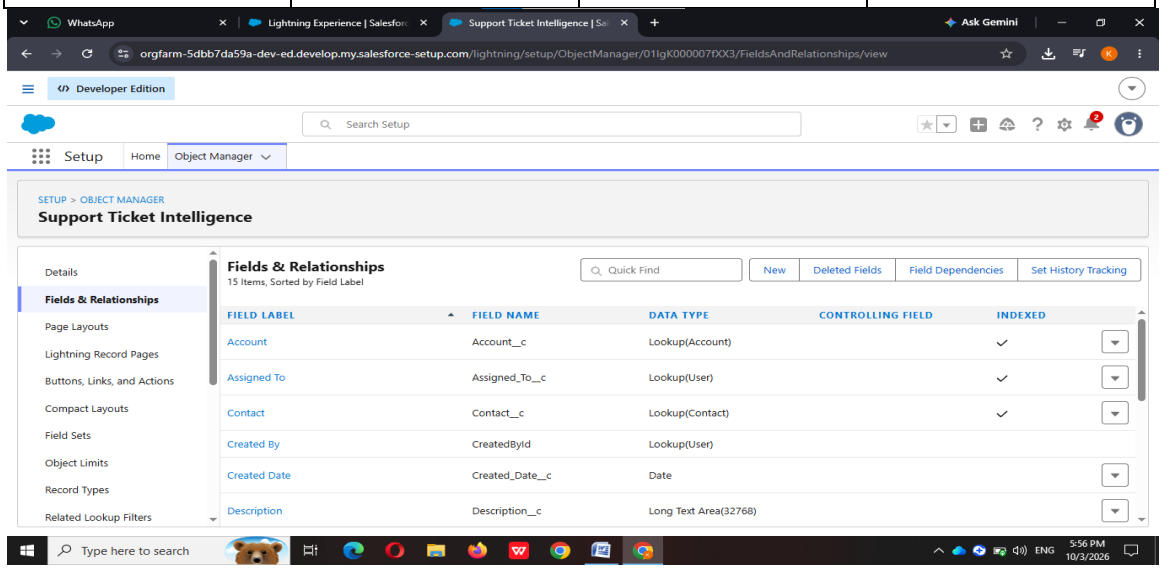
7. Technology Stack

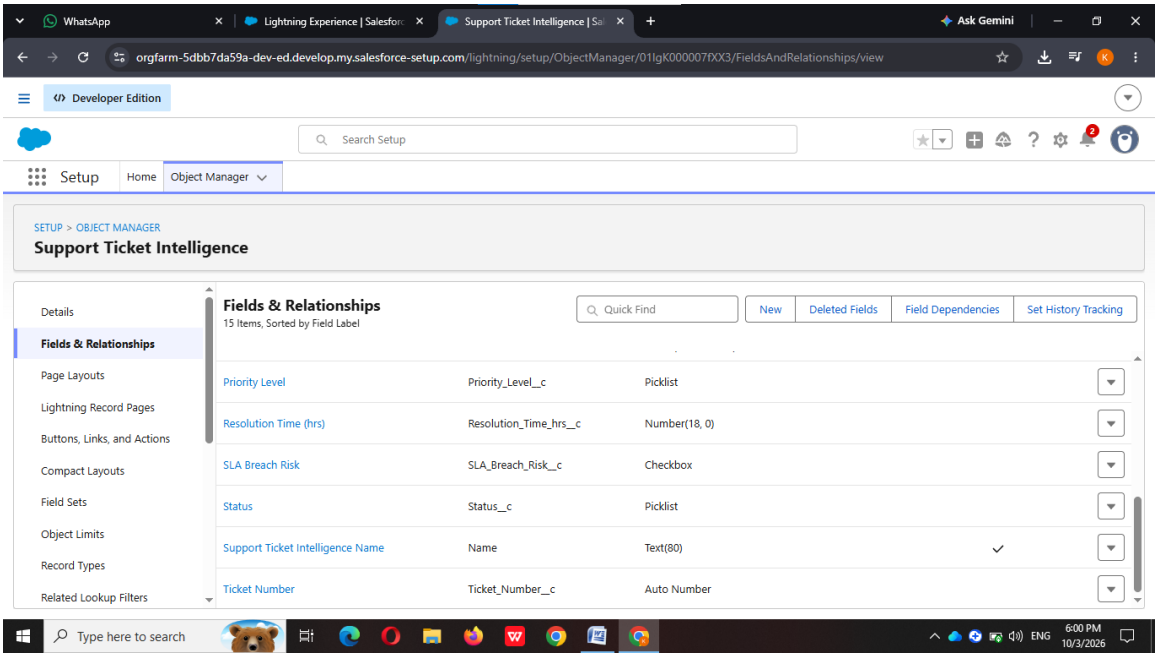
Technology	Purpose
Salesforce	Platform used to build and configure the support ticket system.
Custom Object	Stores support ticket intelligence data.
Salesforce Flow	Performs record retrieval, priority analysis, assignment, and task creation.
Agentforce	Analyzes ticket requests and triggers the configured backend automation.

8. Custom Object and Fields

A custom object named Support Ticket Intelligence was used to store the information needed for the project. Its API name is Support_Ticket_Intelligence__c.

Field Label	API Name	Data Type	Purpose
Ticket Number	Ticket_Number__c	Auto Number	Generates the ticket number.
Customer	Customer__c	Lookup (Account)	Connects the ticket with an Account.
Contact	Contact__c	Lookup (Contact)	Stores the customer contact.
Issue Type	Issue_Type__c	Picklist	Technical, Billing, or General.
Description	Description__c	Long Text Area	Stores the issue details.
Priority Level	Priority_Level__c	Picklist	Low, Medium, or High.
Status	Status__c	Picklist	New, In Progress, or Resolved.
Created Date	Created_Date__c	Date	Stores the ticket date.
Assigned To	Assigned_To__c	Lookup (User)	Stores the support agent.
SLA Breach Risk	SLA_Breach_Risk__c	Checkbox	Indicates the risk flag.
Resolution Time (hrs)	Resolution_Time__c	Number	Stores time taken for resolution.





. **Figure 1. Support Ticket Intelligence Custom Object**

9. Auto-Launched Flow

The project uses an Auto-Launched Flow. It can be called from Agentforce and performs the backend logic required for ticket analysis. The flow receives an account name, finds the related account and latest support ticket, analyzes the description, and prepares the final action message.

9.1 Flow Variables

Variable	Type	Purpose
varAccountName	Text	Input account name.
varAccountId	Text	Stores the account ID.
varTicketId	Text	Stores the ticket ID.
varPriorityLevel	Text	Stores High, Medium, or Low.
varAssignedTo	Text	Stores the assigned support agent.
varActionMessage	Text	Stores the final response message.

9.2 Flow Implementation

Step 1 – Get Account

The flow searches the Account object using the account name and stores the latest matching record.

Step 2 – Store Account ID

The Account ID is stored in varAccountId.

Step 3 – Get Ticket

The flow retrieves the latest Support Ticket Intelligence record associated with the account.

Step 4 – Store Ticket ID

The ticket ID is stored in varTicketId.

Step 5 – Analyze Description

The ticket description is checked for configured keywords.

Step 6 – Set Priority

The flow assigns High, Medium, or Low based on the matching condition.

Step 7 – High Priority Check

The flow checks whether the calculated priority is High.

Step 8 – Create Task

For a High-priority ticket, an Urgent Ticket Handling task is created.

Step 9 – Assign Agent

The configured senior support agent is selected for High-priority handling.

Step 10 – Optional SLA Check

The flow can check whether the ticket is older than two days.

Step 11 – Final Message

The flow prepares a response describing the priority and action taken.

Step 12 – Save Flow

The completed flow is saved and activated.

9.3 Priority Classification

Priority	Configured Keywords
High	urgent, not working, failure
Medium	issue, slow, delay
Low	Used when none of the configured High or Medium keywords are found.

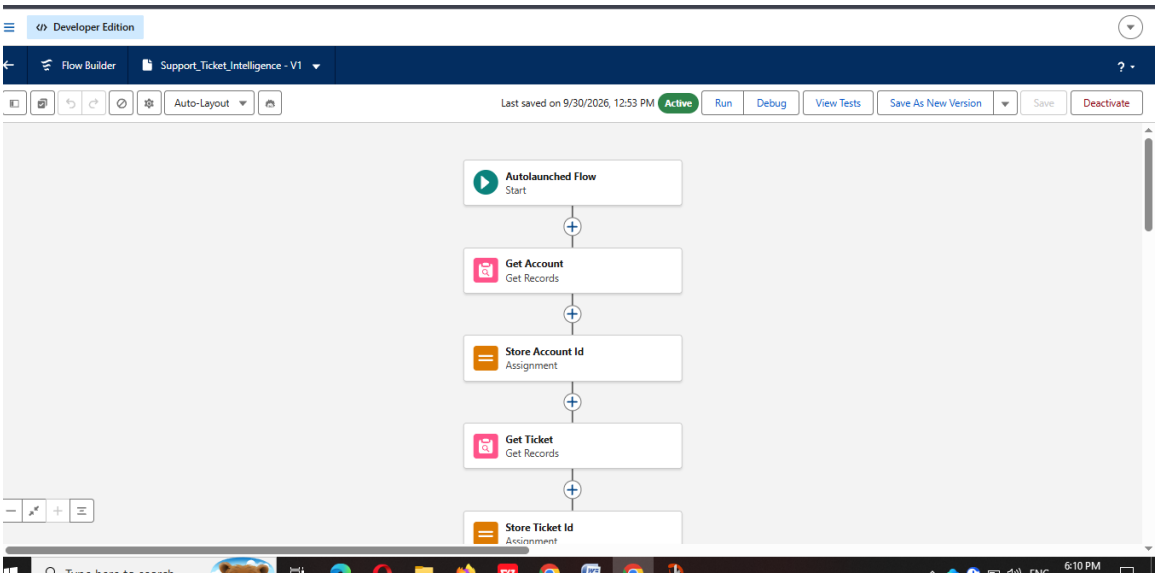


Figure 2. Auto-Launched Flow

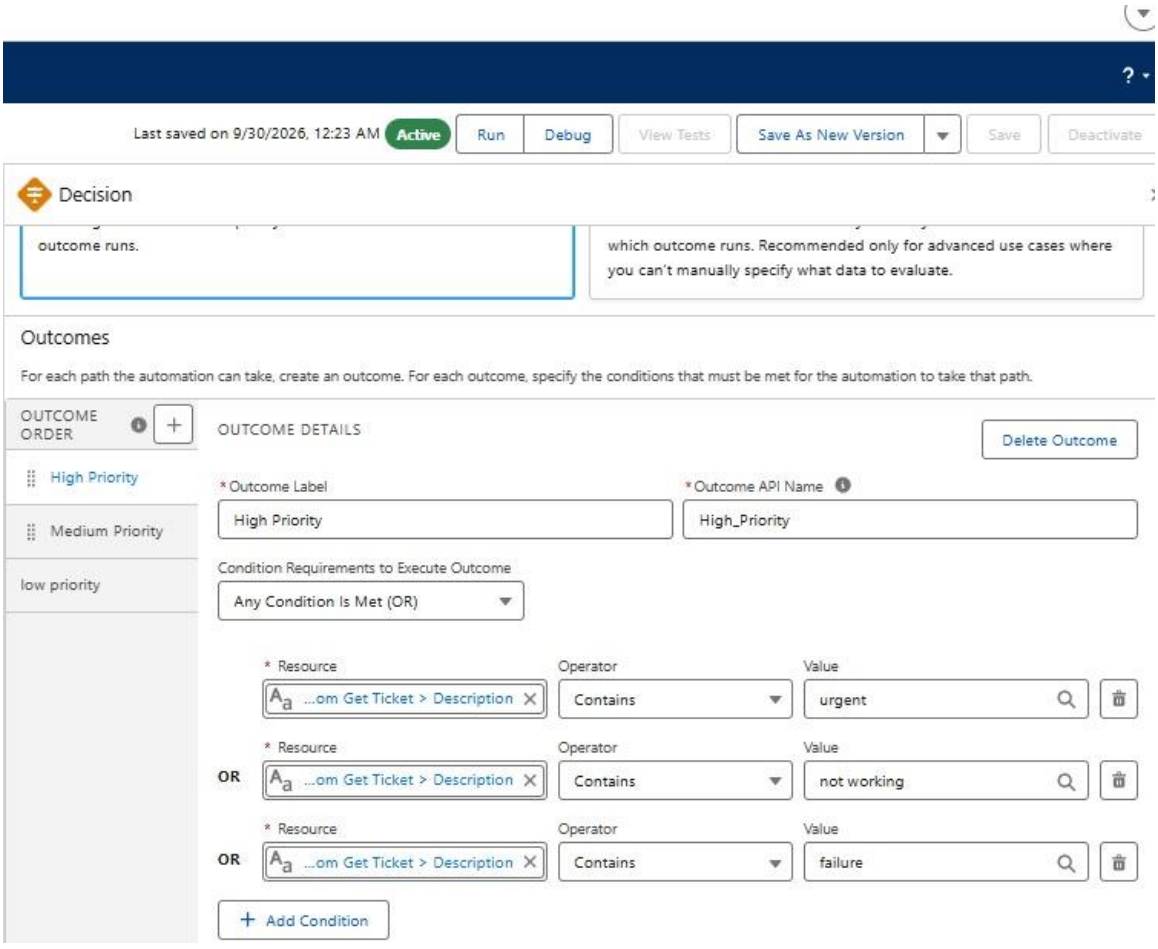


Figure 3. Priority Decision Logic

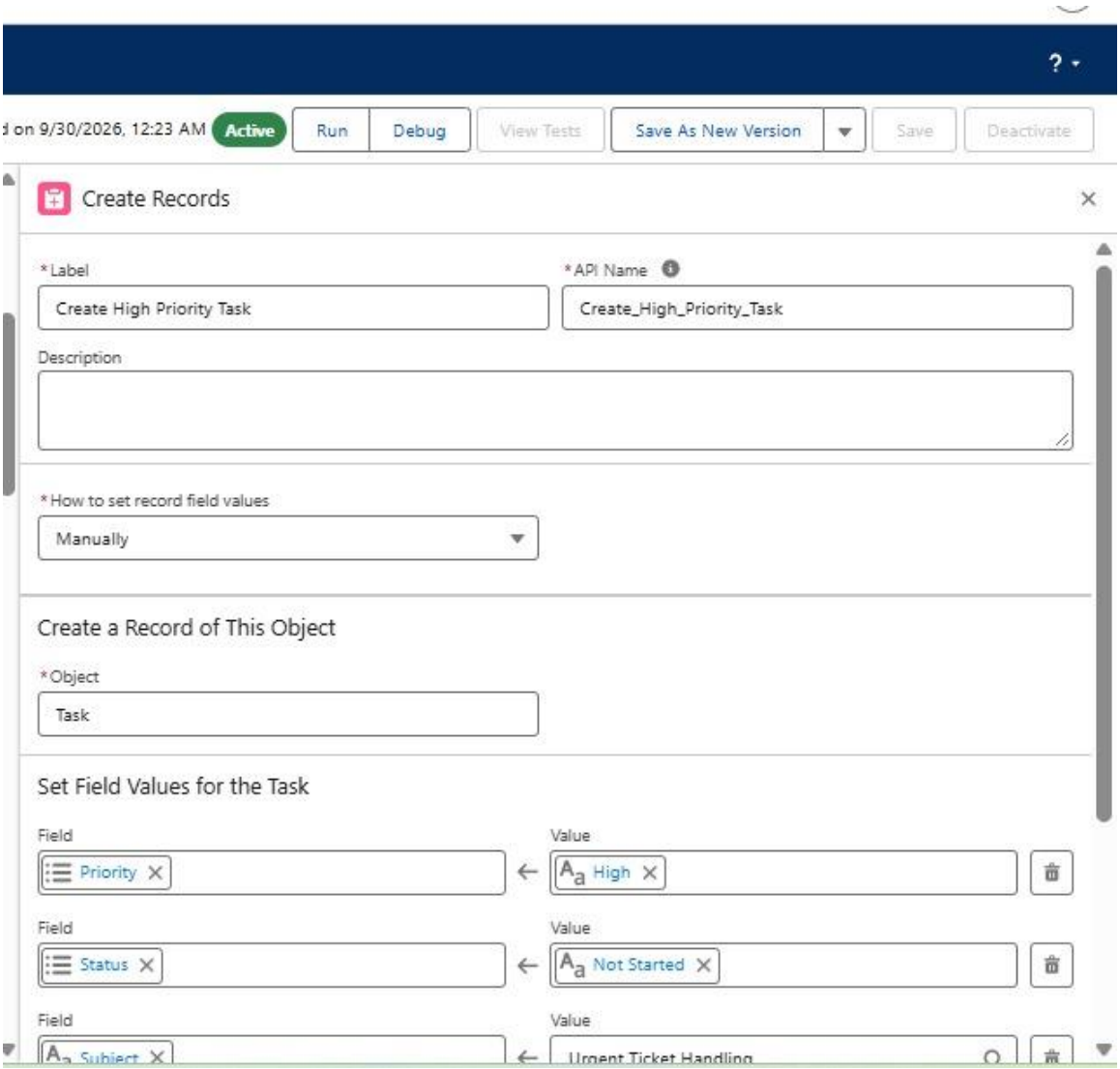


Figure 4. Create Task Configuration

10. Final Flow Structure

The complete process follows a clear sequence: Start → Get Account → Store Account ID → Get Ticket → Store Ticket ID → Analyze Description → Set Priority → Check High Priority → Create Task → Assign Agent → Optional SLA Check → Final Message → End.

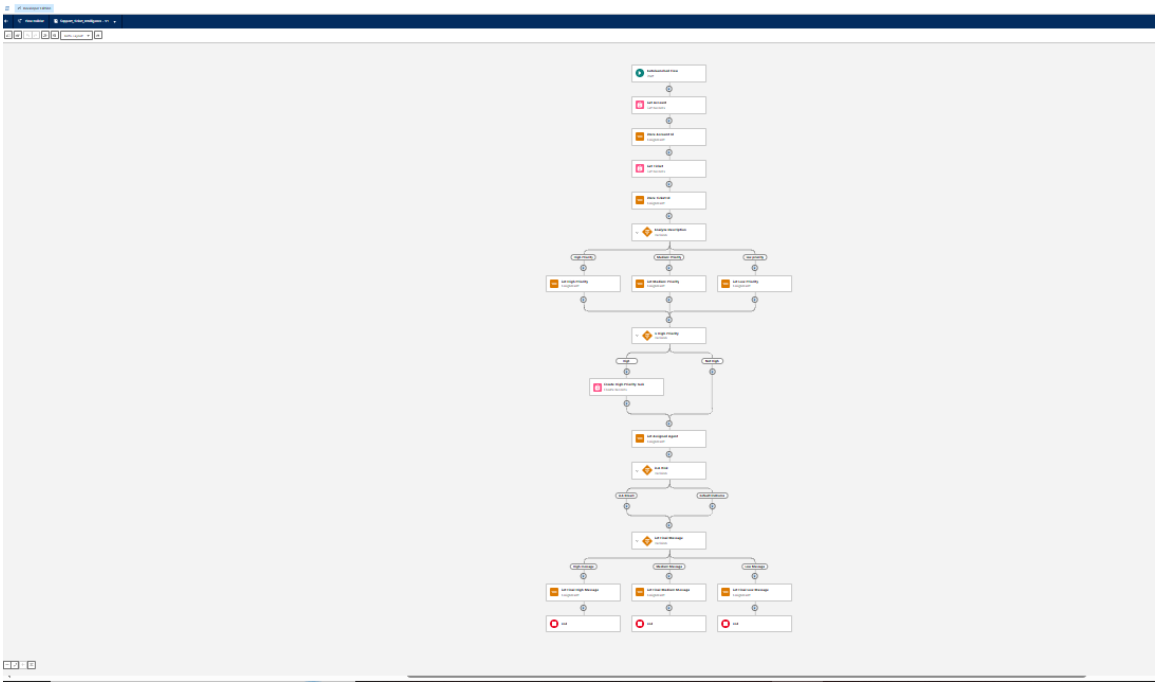


Figure 5. Final Flow Structure

11. Agentforce Configuration

Agentforce is used as the conversational layer for the project. The Agentforce configuration enables the AI agent to receive an account name, retrieve the latest support ticket, analyze the issue description, determine its priority, and trigger the configured automation.

1. Open Salesforce and enable the required Agentforce configuration.

2. Enable the Default Agent.

3. Open the Default Agent in Builder.

4. Create a new subagent.

5. Configure the Support Ticket Priority Analysis topic.

6. Connect the topic with the backend automation.

7. Configure the required input and output variables.

8. Test the final response and output.

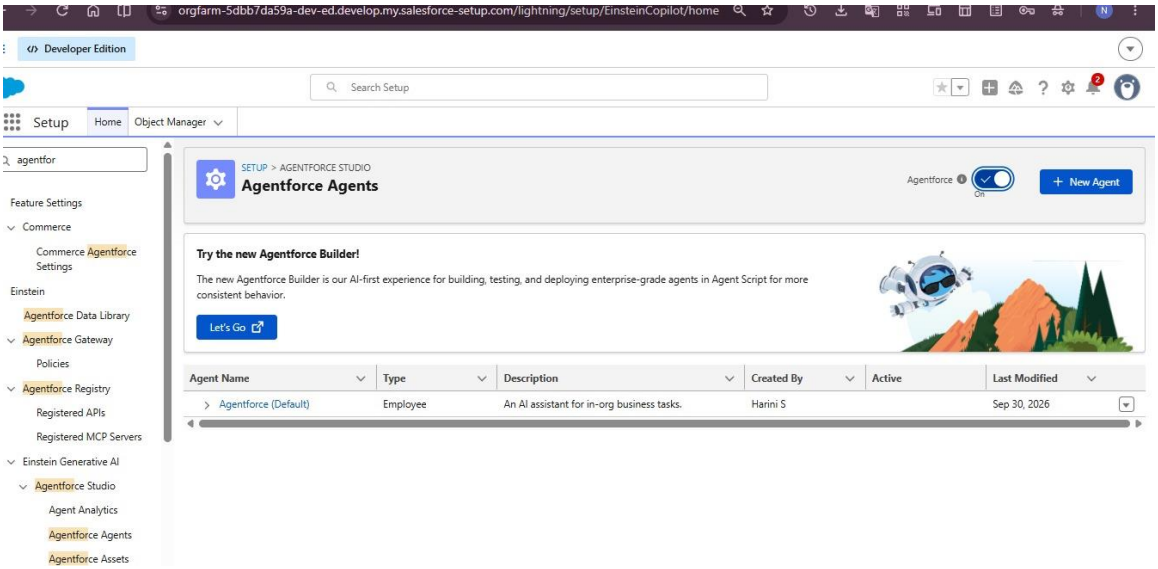


Figure 6. Agentforce Enablement

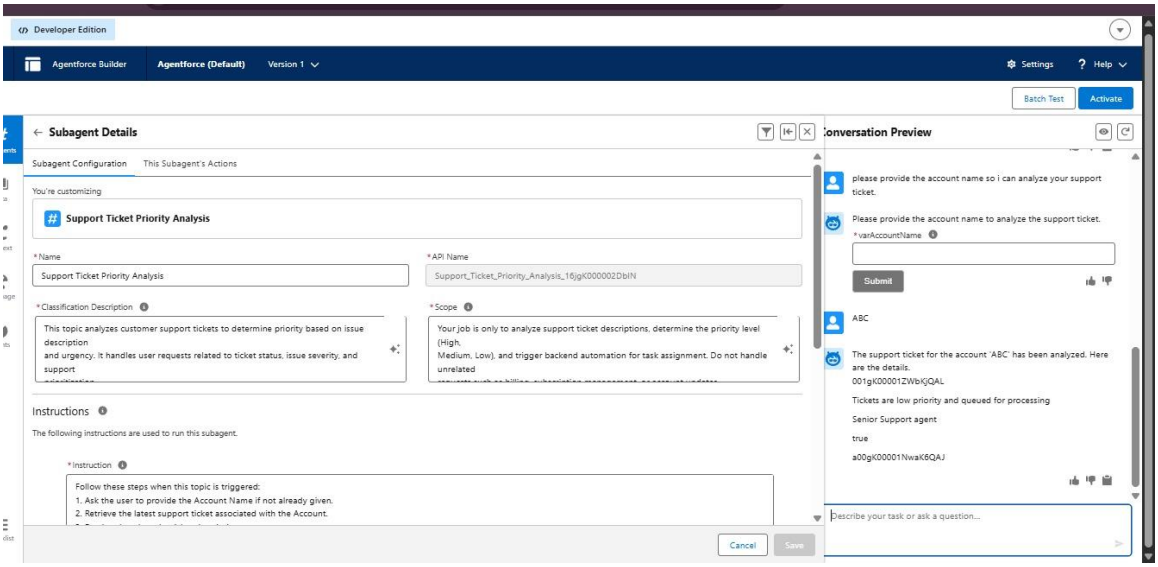


Figure 7. Agentforce Builder

12. Agentforce Subagent Configuration

12.1 Name

Support Ticket Priority Analysis

12.2 Classification Description

This topic analyzes customer support tickets to determine priority based on the issue description and urgency. It handles requests related to ticket status, issue severity, and support prioritization.

12.3 Scope

The agent is limited to analyzing support ticket descriptions, determining High, Medium, or Low priority, and triggering backend automation for task assignment. Unrelated requests such as billing, subscription management, or account updates are outside the defined scope.

12.4 Instructions

1. Ask for the Account Name if it has not already been provided.

2. Retrieve the latest support ticket associated with the account.

3. Read and analyze the ticket description.

4. Determine priority using the configured keywords.

5. Set the priority level.

6. For High priority, trigger the Auto-Launched Flow and ensure an urgent task is created.

7. Assign the ticket to the appropriate support level.

8. Return a clear response to the user.

9. Use Flow for actions rather than updating records directly or sending emails.

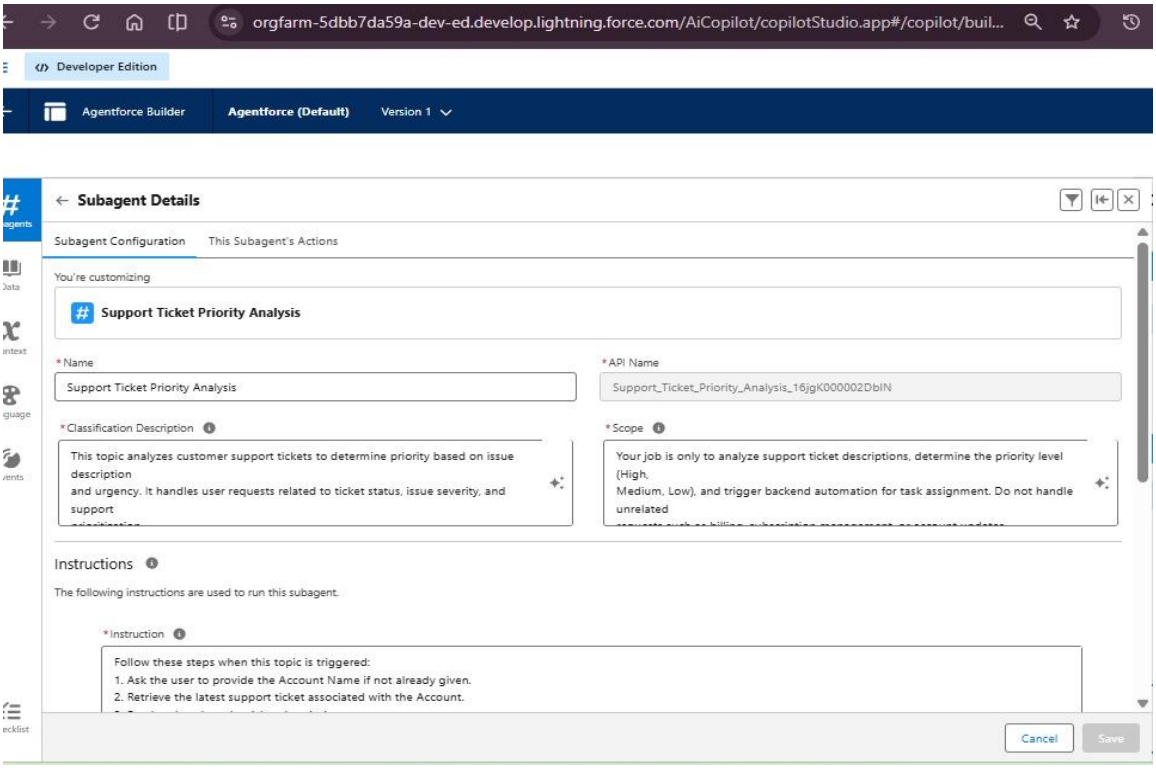


Figure 8. Support Ticket Priority Analysis Subagent

13. Agent Action Configuration

The Agent Action analyzes customer support tickets using the provided account name. It determines the priority from the issue description and triggers the backend automation for high-priority cases.

Variable	Type	Purpose
varAccountName	lightning__textType	Input account name collected from the user.
varAccountId	lightning__textType	Returns the account identifier.
varActionMessage	lightning__textType	Displays the final action message.
varAssignedTo	lightning__textType	Shows the assigned support agent or team.
varPriorityLevel	lightning__textType	Shows the calculated priority.
varTicketId	lightning__textType	Returns the analyzed ticket identifier.

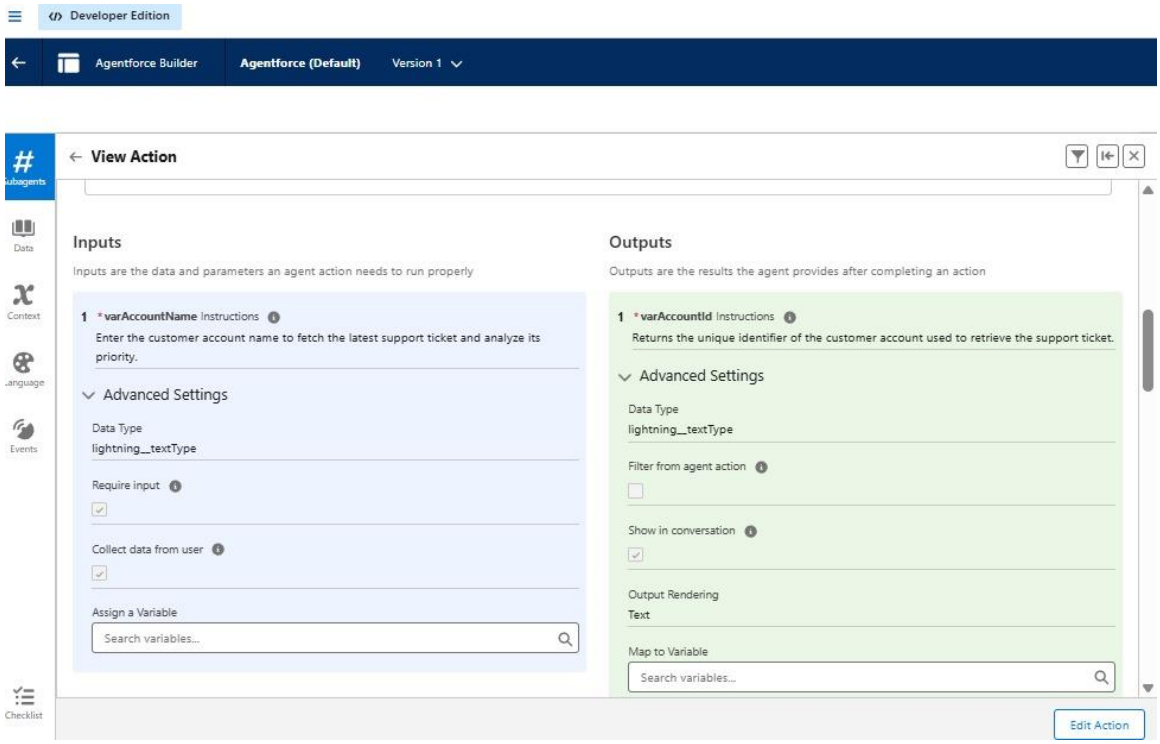


Figure 9. Agent Action Input and Output Configuration

14. System Workflow

The overall workflow begins when the user provides an account name. Agentforce passes this information to the configured action. The Flow finds the account and its latest support ticket, reads the ticket description, and determines the priority. If the ticket is High priority, the system creates an urgent handling task and assigns it to the configured senior support level. Finally, the action message and priority information are returned to Agentforce.

15. Testing and Results

Test Case	Input Condition	Expected Result	Status
High Priority	Description contains urgent/not working/failure	Priority becomes High and urgent task is created.	To be tested
Medium Priority	Description contains issue/slow/delay	Priority becomes Medium.	To be tested
Low Priority	No configured High/Medium keyword	Priority becomes Low.	To be tested
Agentforce Input	Account name provided	Latest related ticket is analyzed.	To be tested

The project document defines the expected automation behavior. Actual test execution results should be updated after the Salesforce configuration is tested with sample records.

16. Challenges Faced and Fixes

Challenge	Approach Used
Identifying the latest ticket	The Flow sorts matching records by Created Date in descending order and stores the first record.
Classifying ticket urgency	Decision conditions use the configured keywords to separate High, Medium, and Low cases.
Reducing manual handling	Agentforce is connected to the backend Flow so the configured automation can be triggered from the agent.
Handling urgent cases	High-priority cases create a dedicated Urgent Ticket Handling task.

17. Conclusion

The Customer Support Ticket Priority Prediction and Automated Assignment System demonstrates how Salesforce, Flow, and Agentforce can be combined to organize customer support ticket handling. The system retrieves the latest ticket, analyzes its description, classifies the priority, and performs additional automation for urgent cases. By reducing repetitive manual steps, the solution provides a more structured way for support teams to focus on important tickets.

18. Appendix – Screenshot Checklist

- Screenshot of Support Ticket Intelligence custom object.
- Screenshot of custom object fields.
- Screenshot of Auto-Launched Flow.
- Screenshot of Get Account configuration.
- Screenshot of Get Ticket configuration.
- Screenshot of priority Decision element.
- Screenshot of Create Task configuration.
- Screenshot of complete Flow structure.
- Screenshot of Agentforce enablement.
- Screenshot of Agentforce Builder.
- Screenshot of Support Ticket Priority Analysis subagent.
- Screenshot of Agent Action configuration.
- Screenshot of final Agentforce output.

Note: Screenshot placeholders have intentionally been included because screenshots were not available at the time of preparing this document. They can be replaced with actual Salesforce screenshots later.